2025 JPS Annual Meeting 19aSK107-1 10:00~

Stabilizer Rényi Entropy and Conformal Field Theory

Masahiro Hoshino
Department of Physics, UTokyo

Masaki Oshikawa
ISSP, UTokyo

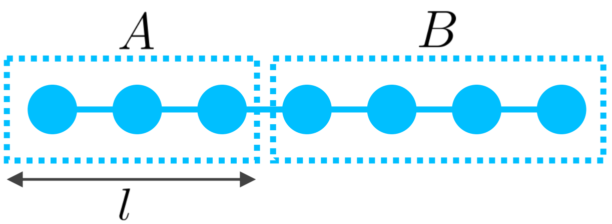
Yuto Ashida
Dept. Phys. & IPI, UTokyo

UTokyo

Scaling of Entanglement Entropy in 1D pure states

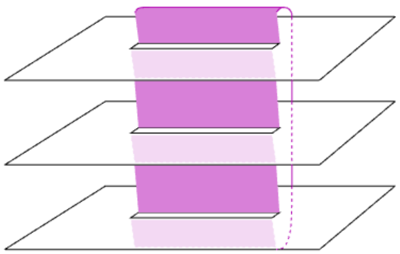
Gapped ground states

M B Hastings, J. Stat. Mech (2007)



- efficient simulation with Matrix Product States (MPS)
- area law $S_A \sim \text{const.}$

Critical (gapless) ground states

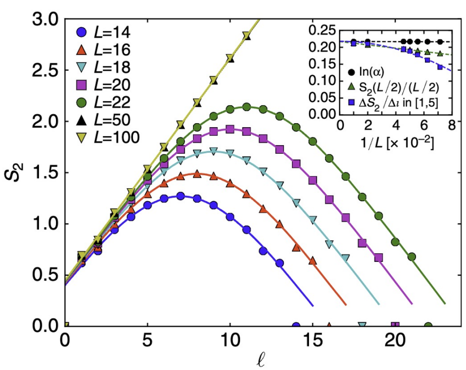


P. Calabrese & J. Cardy, J. Stat. Mech. (2004,2009)

- universal scaling determined by the central charge
- log law $S_A \sim \frac{c}{3} \ln l$

Thermal (typical) pure states

- almost maximally entangled states
- volume law $S_A \sim l$



Y. O. Nakagawa, et al., Nat. Commun. (2018)

EE reflects phases of matter!

Entanglement: resource for quantum communication

Local operations & classical communication (LOCC)

$$|0\rangle|0\rangle \quad \cancel{\longrightarrow} \quad \frac{|0\rangle|0\rangle + |1\rangle|1\rangle}{\sqrt{2}}$$

cannot create entanglement

Separable state

$$\rho = \sum_i p_i \, \rho_i^A \otimes \rho_i^B$$

LOCC

≈quantum channel acting locally

$$\mathcal{E} = \sum_j \mathcal{E}_j^A \otimes \mathcal{E}_j^B$$

Entangled state

$$\rho \neq \sum_i p_i \, \rho_i^A \otimes \rho_i^B$$

Entangling gates & measurements

ex)

$$\mathrm{CNOT} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix} \qquad |\mathrm{Bell}\rangle\langle\mathrm{Bell}|$$

Entanglement entropy (EE) : entanglement measure for pure states

$$S_A = -\mathrm{Tr}[\rho_A \ln \rho_A]$$

$|\psi\rangle$: pure state

$\rho_A = \mathrm{Tr}_B[|\psi\rangle\langle\psi|]$: reduced density matrix

R. Horodecki et al., Rev. Mod. Phys. (2009)

Magic: resource for universal quantum computation

Universal gate set = Clifford + T

$$\mathrm{CNOT} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix} \quad H = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \quad S = \begin{pmatrix} 1 & 0 \\ 0 & i \end{pmatrix}$$

$$T = \begin{pmatrix} 1 & 0 \\ 0 & e^{i\pi/4} \end{pmatrix}$$

Clifford unitary

- easy to implement fault-tolerantly (transversal gates)
- classically simulable due to Gottesman-Knill theorem

T gate

Stabilizer states

(convex hull of) **Clifford** $|0\rangle^{\otimes L}$

Stabilizer protocol

Clifford + other easy operations

Non-stabilizer state (magic state)

$\neq$ **Clifford** $|0\rangle^{\otimes L}$

Non-Clifford gates

ex) T gate, CCZ gate

cf. Eastin-Knill theorem

Magic: resource for universal quantum computation

Universal gate set = Clifford + T

$$\mathrm{CNOT} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix} \quad H = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \quad S = \begin{pmatrix} 1 & 0 \\ 0 & i \end{pmatrix}$$

$$T = \begin{pmatrix} 1 & 0 \\ 0 & e^{i\pi/4} \end{pmatrix}$$

Clifford unitary

- easy to implement fault-tolerantly (transversal gates)
- classically simulable due to Gottesman-Knill theorem

T gate

Example: Bell state $|\mathrm{Bell}^{00}\rangle = \dfrac{|00\rangle + |11\rangle}{\sqrt{2}}$

- **entanglement: resource state**

$$S_A = \ln 2 > 0$$

- **nonstabilizerness: free state**

$$|\mathrm{Bell}^{00}\rangle = \mathrm{CNOT}\, H\, |00\rangle$$

$$H|0\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$$

$$\begin{aligned} &\mathrm{CNOT}(a|00\rangle + b|01\rangle + c|10\rangle + d|11\rangle) \\ &= a|00\rangle + b|01\rangle + c|11\rangle + d|10\rangle \end{aligned}$$

Even highly entangled states can be simulated classically!

Quantifying magic

Amount of magic ≈ distance from stabilizer states

Stabilizer Rényi entropy (SRE): measure for pure states

$$M_\alpha(\psi) = \frac{1}{1-\alpha} \ln \sum_{\vec{m}} \frac{\mathrm{Tr}^{2\alpha}[\sigma^{\vec{m}}\psi]}{2^L}$$

$\psi = |\psi\rangle\langle\psi|$: **L-qubit pure state**

$\sigma^{\vec{m}} = \sigma^{m_1}\sigma^{m_2}\cdots\sigma^{m_L}$: **Pauli string**

L. Leone et al., PRL (2022)

Properties

1. **Faithfulness** : $M_\alpha(\psi) = 0 \iff \text{stabilizer state}$
2. **Additivity** : $M_\alpha(\psi \otimes \phi) = M_\alpha(\psi) + M_\alpha(\phi)$
3. **Monotonicity : non-increasing under stabilizer protocol**
   only for $\alpha \geq 2$

Operational meaning

see also L. Bittel & L. Leone, arXiv:2507.22883

$$\text{Stabilizer protocol}: \ |R_1\rangle^{\otimes M} \to |R_2\rangle^{\otimes N}$$

conversion process

$$\frac{N}{M} \leq \frac{M_{\alpha\geq 2}(R_1)}{M_{\alpha\geq 2}(R_2)}$$

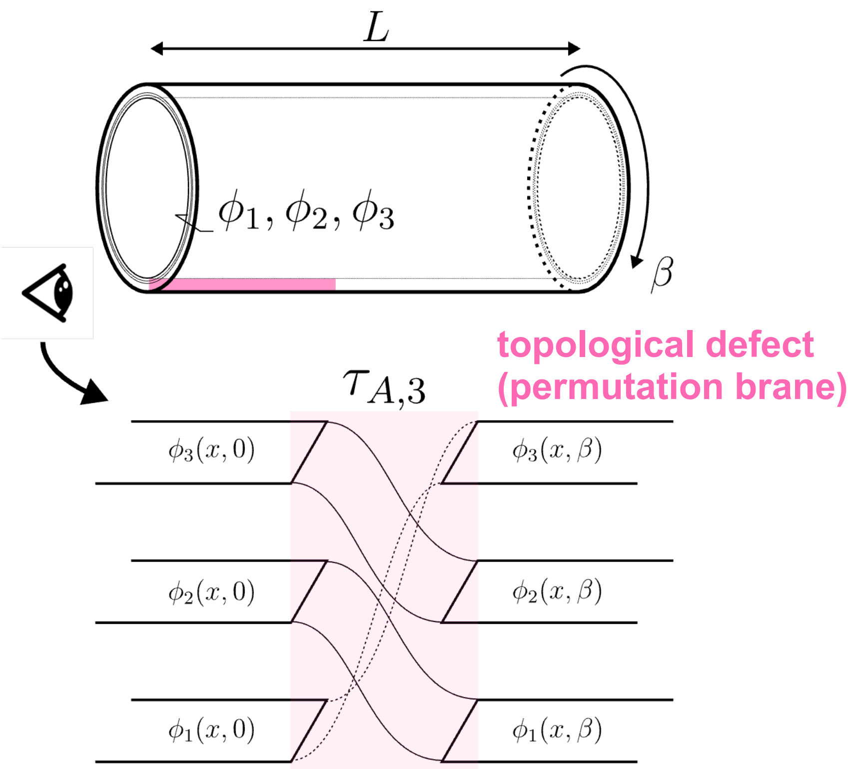
rate of conversion ≤ ratio of SRE

Entanglement entropy vs Stabilizer Rényi entropy

$$S_A^{(\alpha)} = \frac{1}{1-\alpha} \ln \mathrm{Tr}[\rho_A^{\alpha}] \quad (\rho_A = \mathrm{Tr}_{\bar{A}}\, \psi)$$
$$= \frac{1}{1-\alpha} \ln \mathrm{Tr}\left[\tau_{A,\alpha} \psi^{\otimes \alpha}\right]$$

Permutation operator
invariant under arbitrary unitaries
∵ Schur-Weyl duality

$$Z_3 = \mathrm{Tr}\left[\tau_{A,3}(e^{-\beta H})^{\otimes 3}\right]$$

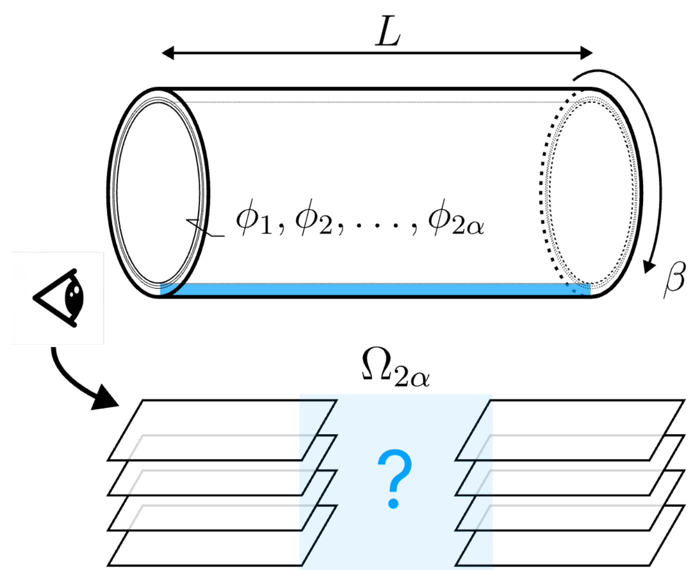


$$M_\alpha = \frac{1}{1-\alpha} \ln \mathrm{Tr}\left[\Omega_{2\alpha} \psi^{\otimes 2\alpha}\right]$$
$$\left(\Omega_{2\alpha} = \frac{1}{2^L} \sum_{\vec{m}} (\sigma^{\vec{m}})^{\otimes 2\alpha}\right)$$

Clifford commutant
invariant under arbitrary *Clifford* unitaries

L. Bittel, et al., arXiv:2504.12263

$$Z_{2\alpha} = \mathrm{Tr}\left[\Omega_{2\alpha}(e^{-\beta H})^{\otimes 2\alpha}\right]$$



Calculation reduces to determining the line defect!

Replica trick + Folding trick → Boundary CFT

Replica trick

$$M_\alpha(\psi) = \frac{1}{1-\alpha} \ln \sum_{\vec{m}} \mathrm{Tr}^\alpha [P^{\vec{m}}(\psi \otimes \psi^*)] - (\ln 2)L$$

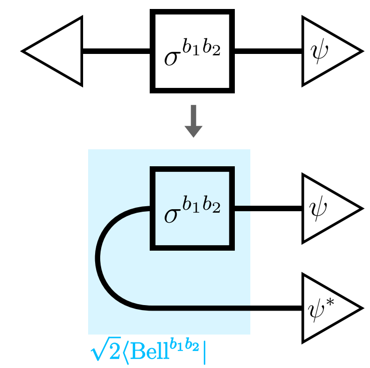
$$= \frac{1}{1-\alpha} \ln \mathrm{Tr} \left[\sum_{\vec{m}} (P^{\vec{m}})^{\otimes\alpha} (\psi \otimes \psi^*)^{\otimes\alpha} \right] - (\ln 2)L$$

projection operator (rank > 1)

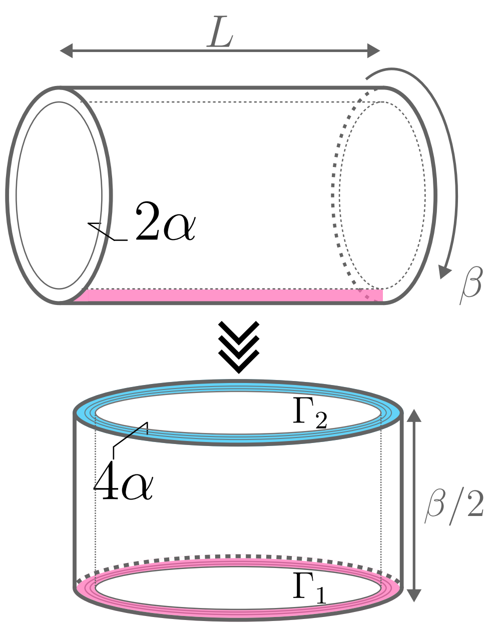
partition function of 2α-component CFT
with a boundary perturbation $\sum_{\vec{m}} (P^{\vec{m}})^{\otimes\alpha} \to e^{-\delta \mathcal{S}}$

Choi-Jamiolkowski isomorphism

$$\mathrm{Tr}\left[(\sigma^{\vec{m}})^{\otimes 2} \psi^{\otimes 2}\right] = 2^L\, \mathrm{Tr}\left[P^{\vec{m}}(\psi \otimes \psi^*)\right]$$



Folding trick



$$\frac{Z_{2\alpha}}{Z^{2\alpha}} = \mathrm{Tr} \left[\sum_{\vec{m}} (P^{\vec{m}})^{\otimes\alpha} (\psi \otimes \psi^*)^{\otimes\alpha} \right]$$

torus of size $L \times \beta$

cylinder of circumference L, length $\beta/2$
H: Hamiltonian of 4α-component CFT

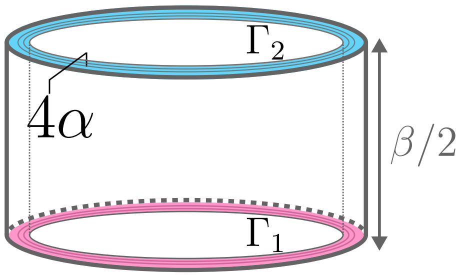
$$Z_{2\alpha} = \langle \Gamma_2 | e^{-\frac{\beta}{2} H} | \Gamma_1 \rangle$$

boundary state from the line defect

boundary state from the folding

Result: SRE contains a universal constant term

Noninteger ground-state degeneracy (g-factor)



$$Z_{2\alpha} = \langle \Gamma_2 | e^{-\frac{\beta}{2}H} | \Gamma_1 \rangle \underset{\beta\to\infty}{\sim} \langle \Gamma_2 | \mathrm{GS} \rangle \langle \mathrm{GS} | \Gamma_1 \rangle e^{-\frac{\beta}{2}E_{\mathrm{GS}}(L)}$$

Free energy

$$-\ln \frac{Z_{2\alpha}}{Z^{2\alpha}} = bL - \ln g_1 g_2 + o(1)$$

$$g_1 = \langle \mathrm{GS} | \Gamma_1 \rangle$$

$$g_2 = \langle \mathrm{GS} | \Gamma_2 \rangle = 1$$

Stabilizer Rényi Entropy

$$M_\alpha(\psi) = \frac{1}{1-\alpha} \ln \frac{Z_{2\alpha}}{Z^{2\alpha}} - (\ln 2)L = m_\alpha L - c_\alpha + o(1)$$

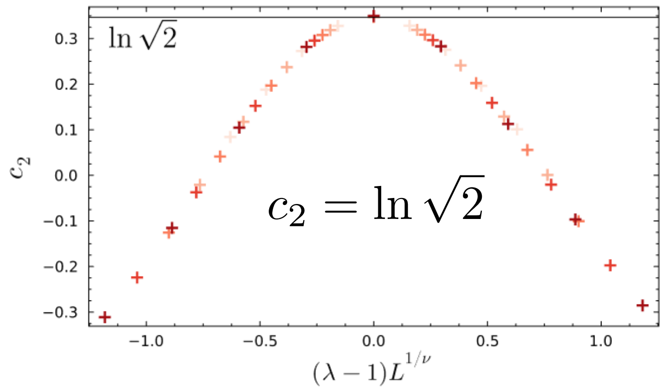
$$c_\alpha = \frac{\ln g_1}{\alpha - 1}$$

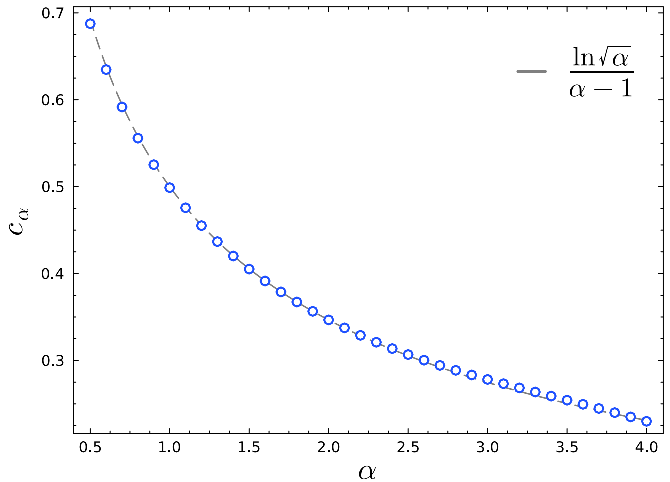
universal constant term derived from the g-factor of Γ_1

Example: Ising model

$$H_{\mathrm{Ising}} = -\sum_{j=1}^{L} (Z_j Z_{j+1} + \lambda X_j)$$

g-factor $g_1 = \langle \mathrm{GS} | \Gamma_1 \rangle = \sqrt{\alpha}$





How to determine the line defect (boundary condition)

1. Boundary perturbation approach

$$\sum_{b_1,b_2}(P^{b_1b_2})^{\otimes\alpha} = \prod_{s\in\tilde{S}^{(\alpha)}} \frac{I+s}{2} \propto \lim_{\mu\to\infty} e^{\mu\sum s}$$

$$\tilde{S}^{(\alpha)} = \{\underbrace{XX^{(1)}XX^{(2)}}, \underbrace{ZZ^{(1)}ZZ^{(2)}}, \ldots, \underbrace{ZZ^{(\alpha-1)}ZZ^{(\alpha)}}\}$$

interlayer coupling

$$\Longrightarrow \quad \delta\mathcal{S}_{2\alpha} = -\mu\int d\tau\,\delta(\tau)\int dx \sum_{s\in\tilde{S}^{(\alpha)}} s$$

should be expressed by operators in the CFT
possibly induces *irrelevant* RG flow

2. Magic entropy route

entanglement created by convolution operation $\boxtimes_K\rho := \mathrm{Tr}_{2,3,\ldots,K}[V(\underbrace{\rho\otimes\cdots\otimes\rho}_{K})V^\dagger]$

$$\mathrm{ME}_n^{(K)}(\psi) := S_n(\boxtimes_K\psi) \quad \left(S_n(\rho) = \frac{1}{1-n}\ln\mathrm{Tr}\rho^n\right)$$

equivalent to SRE

$$M_\alpha(\psi) = (\alpha-1)^{-1}\mathrm{ME}_2^{(\alpha)}(\psi)$$

$$\mathrm{ME}_2^{(K)}(\psi) = -\ln\mathrm{Tr}\left[\sum_{\vec{x}_1,\vec{y}_1} P(\vec{x}_1,\vec{y}_1)(\psi\otimes\psi^*)^{\otimes\alpha}\right]$$

$\vec{x}_i$ **computational basis of L-qubit system**

$$\vec{x}_{\mathrm{CM}} := \sum_{j=1}^{K}\vec{x}_j$$

projection operator on

$$|\Psi(\vec{x}_1,\vec{y}_1)\rangle = (V^\dagger\otimes V^\top)|\vec{x}_1,\vec{y}_1\rangle|\underbrace{\Phi\cdots\Phi}_{K-1}\rangle = 2^{-L(K-1)/2}\sum_{\Gamma}|\vec{x}_{\mathrm{CM}}\vec{y}_{\mathrm{CM}}\rangle\otimes_{i=2}^{K}|\vec{x}_{\mathrm{CM}}-\vec{x}_i\;\vec{y}_{\mathrm{CM}}-\vec{y}_i\rangle$$

maximally entangled states

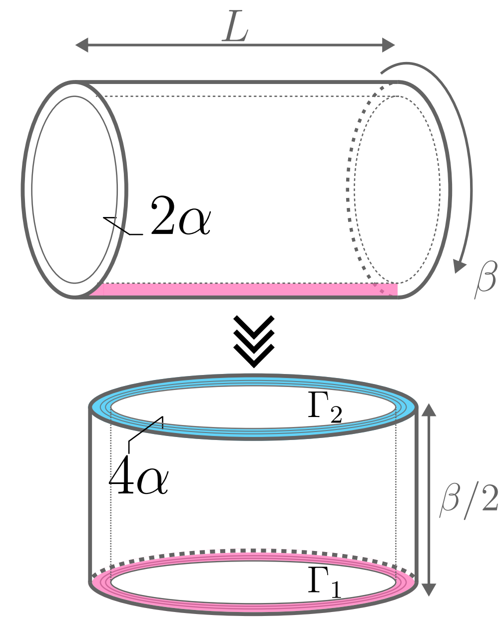
$$\Gamma:\ \vec{x}_{\mathrm{CM}}-\vec{x}_i = \vec{y}_{\mathrm{CM}}-\vec{y}_i \quad (i=2,3,\ldots,K)$$

lattice configuration → line defect in CFT

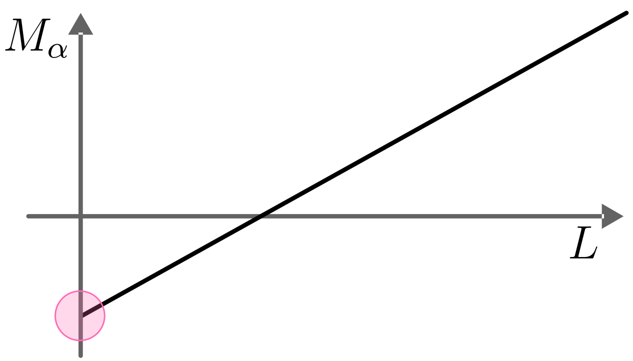
Summary

Nonlinear function of the density matrix (typical in informational quantity)

- **replica trick & folding trick: express the partition function as amplitudes between boundary states of the multicomponent theory**



Universal behavior of the stabilizer Rényi entropy



$$c_\alpha = \frac{\ln g_1}{\alpha - 1}$$

universal constant term derived from the g-factor of Γ_1

$$M_\alpha(\psi) = m_\alpha L - c_\alpha + o(1)$$

Papers

1. **MH, M. Oshikawa, Y. Ashida, arXiv:2503.13599**
2. **MH, Y. Ashida, arXiv:2507.10656**

Appendix

Example: Ising model

$$H_{\mathrm{Ising}} = -\sum_{j=1}^{L} (Z_j Z_{j+1} + \lambda X_j)$$

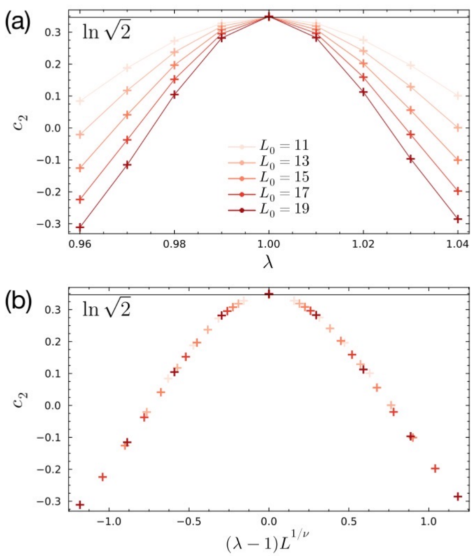
g-factor $\quad g_1 = \langle \mathrm{GS} | \Gamma_1 \rangle = \sqrt{\alpha}$

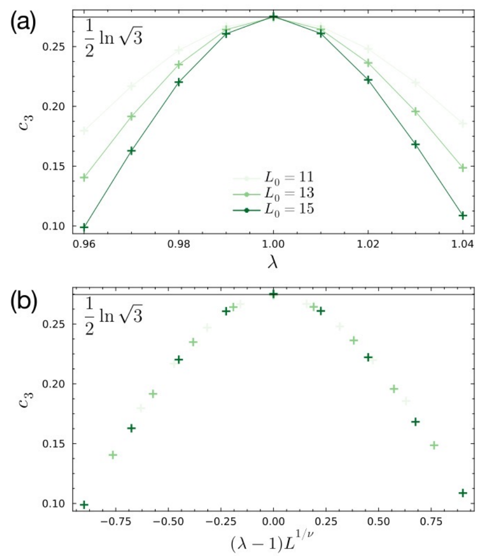
universal constant term

$$c_\alpha = \frac{\ln \sqrt{\alpha}}{\alpha - 1}$$

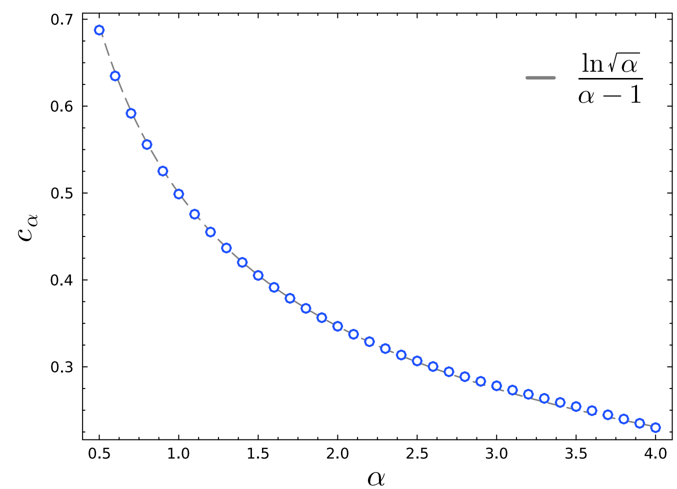
see MH, M. Oshikawa, Y. Ashida, arXiv:2503.13599 for derivation

$$c_2 = \ln \sqrt{2} \qquad\qquad c_3 = \frac{1}{2} \ln \sqrt{3}$$





replica Pauli MPS method **Tarabunga et al., PRL 133, 010601 (2024)**



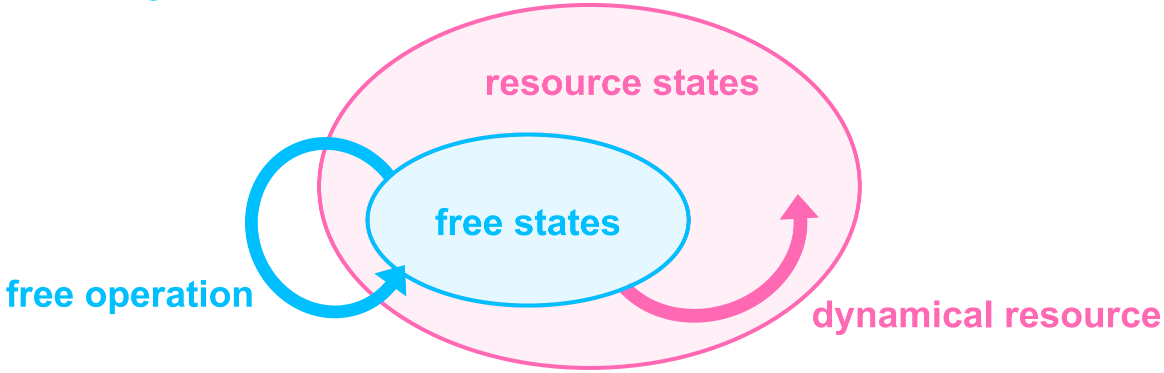
perfect Pauli sampling method **G. Lami & M. Collura, PRL 131, 180401 (2023)**

General framework: quantum resource theory

Resource ≈ things that cannot be easily obtained

example: entanglement, nonstabilizerness (magic), non-Gaussianity, quantum coherence, …

Free states	**Resource states**
restricted class of states (easily prepared)	states that are not free
Free operations	**Dynamical resources**
operations acting invariantly on free states	operations that are not free
easy	hard



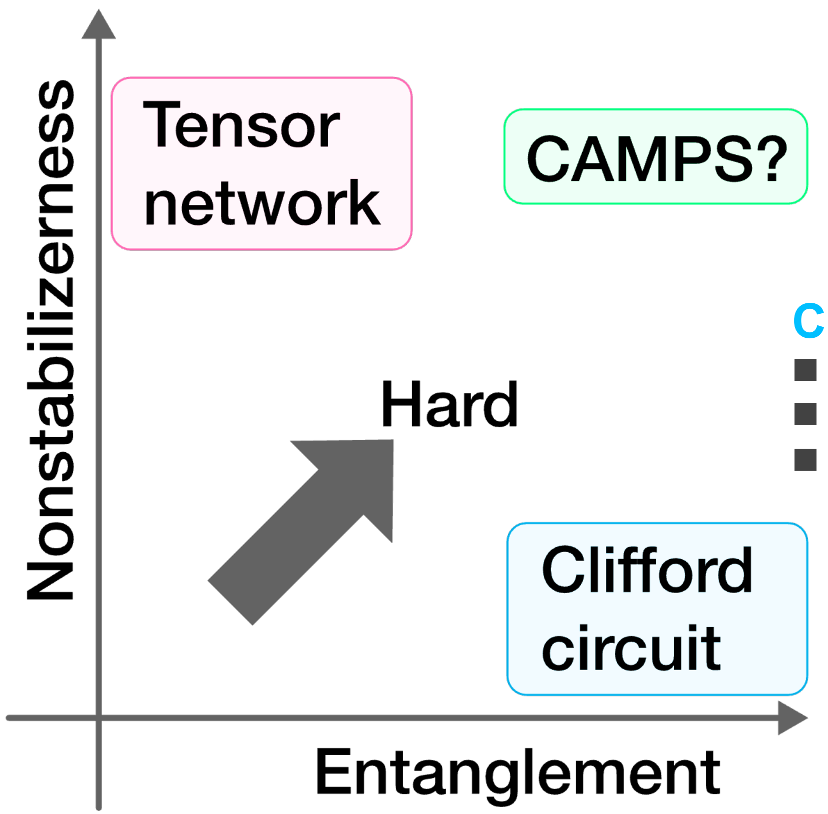
E. Chitambar & G. Gour, Rev. Mod. Phys. (2019)

Hardness of classical simulation

Entanglement alone does not indicate “quantumness”

Tensor network

- able to simulate low-entangled states
- cost $\approx$ bond dimension χ
- scales as $\chi \propto \exp(S_A)$



Clifford circuit

- able to simulate low-magic states
- cost $\approx$ number of T-gates
- scales as $\mathrm{time} \sim \mathrm{poly}(n, m) + \exp(T)$
  (n: # of qubits, m: # of Clifford gates, T: # of T-gates)

Bravyi & Gosset, PRL (2016) / arXiv:1601.07601

Gottesman-Knill theorem: stabilizer protocols can be simulated classically

Stabilizer formalism

$$\sigma^{00} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \ \sigma^{01} = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

$$\sigma^{10} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \ \sigma^{11} = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}$$

stabilizer state $|\psi\rangle$

stabilizer group $S = \{P \in \mathcal{P}_n \mid P|\psi\rangle = |\psi\rangle\}$

Pauli group: n-fold tensor products of Pauli matrices

generators $S = \langle g_1, g_2, \ldots, g_n \rangle$

stabilizer states are completely determined by their generators

1. **Clifford unitary: maps Pauli string to Pauli string**

$$H\sigma^{b_1 b_2} H^\dagger = \sigma^{b_2 b_1} \qquad S\sigma^{b_1 b_2} S^\dagger = \sigma^{b_1 (b_1 \oplus b_2)}$$

$$\mathrm{CNOT}\underbrace{\sigma^{b_1 b_2}}_{\text{control}}\underbrace{\sigma^{b_1' b_2'}}_{\text{target}}\mathrm{CNOT}^\dagger = \sigma^{b_1 (b_2 \oplus b_2')}\sigma^{(b_1 \oplus b_1') b_2'} \qquad b_1 b_2 \in \{0,1\}^2$$

(with possible phase factor)

2. **measurement in the computational basis**

$$|0\rangle\langle 0| = \frac{I+Z}{2}$$

$$|1\rangle\langle 1| = \frac{I-Z}{2} \qquad (Z = \sigma^{01})$$

measure the j-th qubit

- Z_j **commutes with** $g_1, g_2, \ldots, g_n$ $\quad\rightarrow\quad \langle g_1, g_2, \ldots, g_n \rangle$
- Z_j **anti-commutes with** g_k $\quad\rightarrow\quad \langle g_1, \ldots, \underset{Z_j}{\cancel{g_k}}, \ldots, g_n \rangle$

Gottesman-Knill theorem: Clifford can be simulated classically

Stabilizer formalism

$$\sigma^{00} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix},\ \sigma^{01} = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

$$\sigma^{10} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix},\ \sigma^{11} = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}$$

stabilizer state $|\psi\rangle$

stabilizer group $S = \{P \in \mathcal{P}_n \mid P|\psi\rangle = |\psi\rangle\}$

Pauli group: n-fold tensor products of Pauli matrices

generators $S = \langle g_1, g_2, \ldots, g_n \rangle$

Example $|\mathrm{Bell}^{00}\rangle = \dfrac{|00\rangle + |11\rangle}{\sqrt{2}}$

stabilizer group $S = \langle X_1 X_2, Z_1 Z_2 \rangle$ $\because$

$$Z_1 Z_2 |\mathrm{Bell}^{00}\rangle = |\mathrm{Bell}^{00}\rangle$$

$$X_1 X_2 |\mathrm{Bell}^{00}\rangle = |\mathrm{Bell}^{00}\rangle$$

projection operator $|\mathrm{Bell}^{00}\rangle\langle\mathrm{Bell}^{00}| = \dfrac{I + X_1 X_2}{2} \dfrac{I + Z_1 Z_2}{2}$

measurement in the first qubit ($X_1 X_2$ anti-commutes with Z_1)

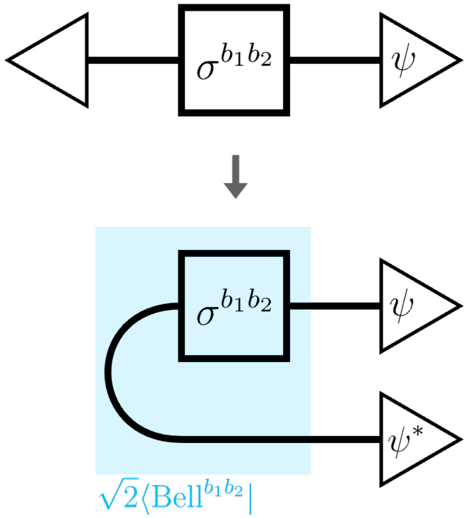
- **outcome +1** $\rightarrow \langle Z_1, Z_1 Z_2 \rangle$ **post-measurement state** $|00\rangle$
- **outcome -1** $\rightarrow \langle -Z_1, Z_1 Z_2 \rangle$ **post-measurement state** $|11\rangle$

highly entangled states can still be simulated efficiently

SRE as participation entropy in the Bell basis

Choi-Jamiolkowski isomorphism

$$\mathrm{Tr}\left[(\sigma^{\vec{m}})^{\otimes 2}\psi^{\otimes 2}\right] = 2^L\,\mathrm{Tr}\left[P^{\vec{m}}(\psi\otimes\psi^*)\right]$$



Participation entropy

$$\mathrm{PE} = \frac{1}{1-\alpha}\ln\sum_i p_i^\alpha \quad (p_i = |\langle i|\Psi\rangle|^2)$$

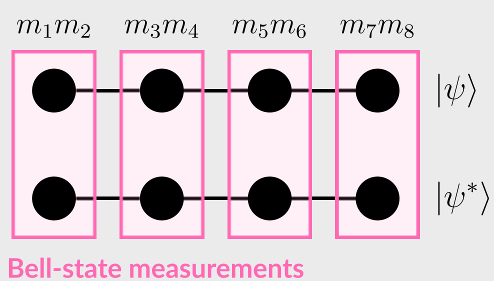
- classical entropy in an orthonormal basis
- Well-explored for 1D CFT in computational basis

Stephan et al., PRB (2009)
Ashida, Furukawa, Oshikawa, PRB (2024)
…

SRE = participation entropy in the Bell basis

$$M_\alpha(\psi) = \frac{1}{1-\alpha}\ln\sum_{\vec{m}}\underbrace{\mathrm{Tr}^\alpha[P^{\vec{m}}(\psi\otimes\psi^*)]}_{p_{\vec{m}}^\alpha} - (\ln 2)L$$

- constant offset to ensure faithfulness
- PE with $|i\rangle = \bigotimes_{j=1}^{L}|\mathrm{Bell}^{m_{2j-1}m_{2j}}\rangle$ and $|\Psi\rangle = |\psi\rangle|\psi^*\rangle$

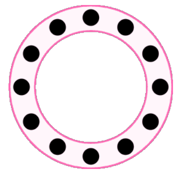


see also: Haug and Kim, PRXQ (2023)

Quantum magic in many-body systems

Full state SRE

$$M_\alpha(\psi) = m_\alpha L - c_\alpha$$



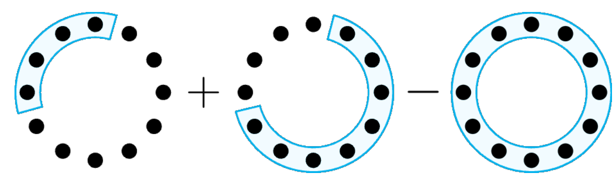
Turkeshi, et al., PRB 111, 054301, (2025)

- **Harr random:** $c_2 = 2,\ c_{>2} = 0$
- **typical state with time-reversal symmetry:** $c_2 = \ln 7,\ c_{>2} = 0$

for critical states, c_α reflect universal data of the underlying CFT

Mutual SRE

$$W_\alpha(A:B) = M_\alpha(\rho_A) + M_\alpha(\rho_B) - M_\alpha(\rho_{AB})$$



◆ **logarithmic scaling for CFT, coefficient determined by the scaling dimension of a BCCO**

$$W_\alpha(l) \sim \ln l \quad \left(l \to \frac{L}{\pi}\sin\frac{\pi l}{L}\right)$$ for periodic chains

Piemontese et al., PRB (2023)
Tarabunga et al., PRXQ (2023), Quantum (2024)
Falcao et al., PRB (2025)
Ding et al., arXiv:2501.12146
…

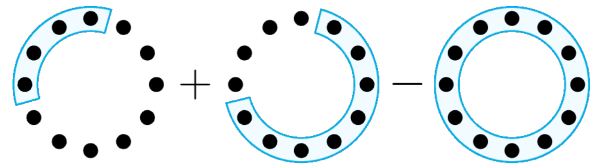
◆ **related to entanglement that can be reduced by Clifford circuits**

Frau, et al., arXiv:2411.11720

BCFT: boundary condition changing operator

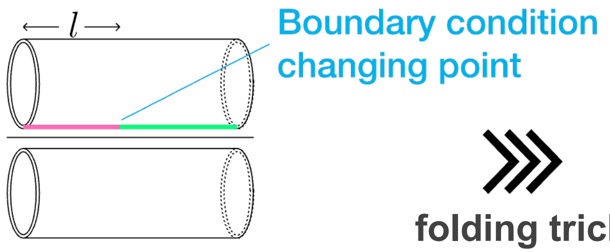
Mutual SRE

$$W_\alpha(A:B) = M_\alpha(\rho_A) + M_\alpha(\rho_B) - M_\alpha(\rho_{AB})$$

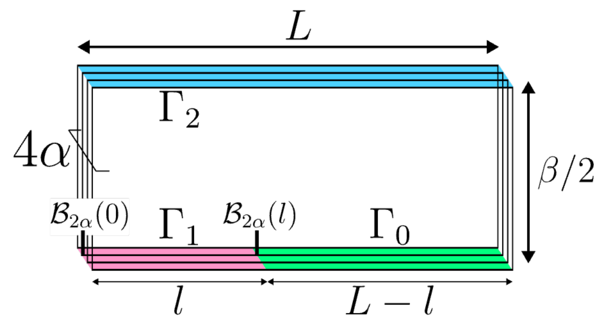


$A = \{1, 2, \ldots, l\}$ $\quad$ $B = \{l+1, l+2, \ldots, L\}$

$$W_\alpha(l) = \frac{2}{1-\alpha} \ln$$



folding trick



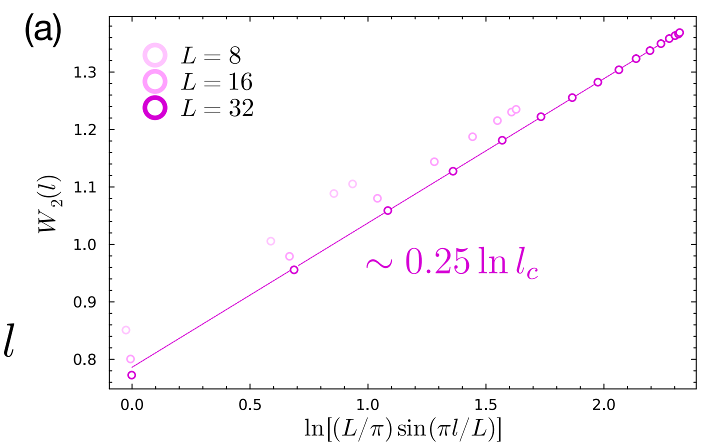
$$Z_{2\alpha}(A) = \langle \mathcal{B}_{2\alpha}(0)\mathcal{B}_{2\alpha}(l)\rangle \sim l^{-2\Delta_{2\alpha}}$$

two-point function of BCCO

Universal logarithmic scaling

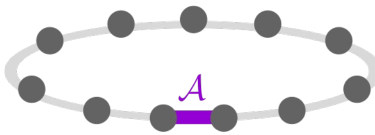
$$W_\alpha(l) \sim \frac{4\Delta_{2\alpha}}{\alpha - 1} \ln l$$

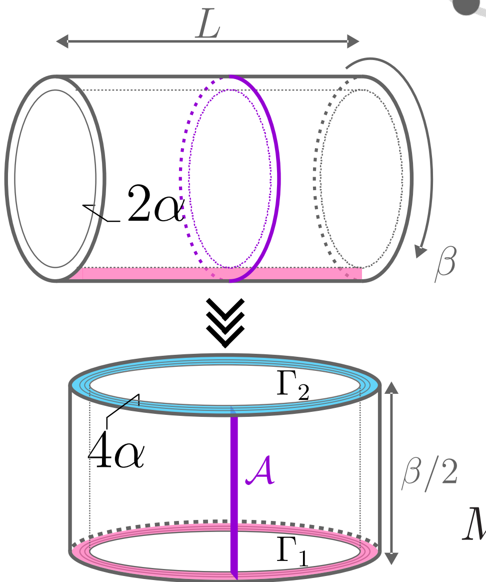
$$\Delta_{2\alpha} = \frac{1}{16}$$ **for the Ising CFT** $$W_2(l) = \frac{1}{4}\ln l$$



Stabilizer Rényi entropy with conformal defects

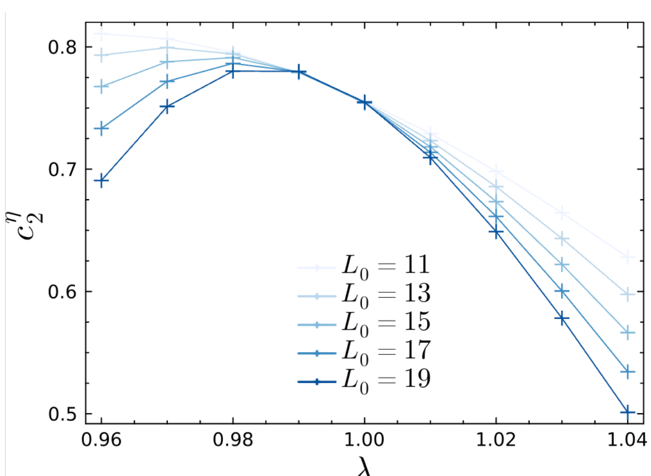
Topological defect (totally transmissive)



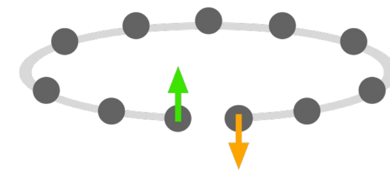


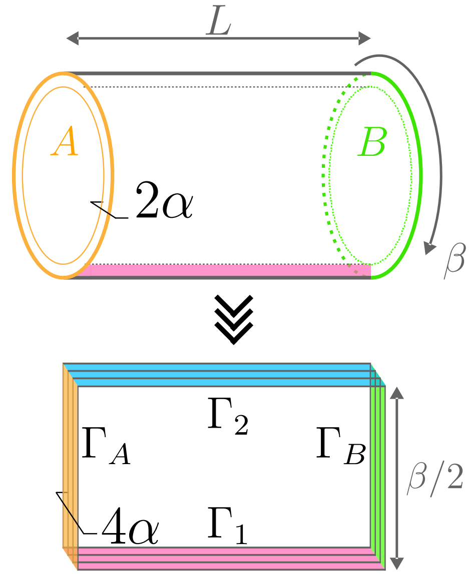
$$M_\alpha = m_\alpha L - c_\alpha^{\mathcal{A}}$$

different bulk sector different constant term



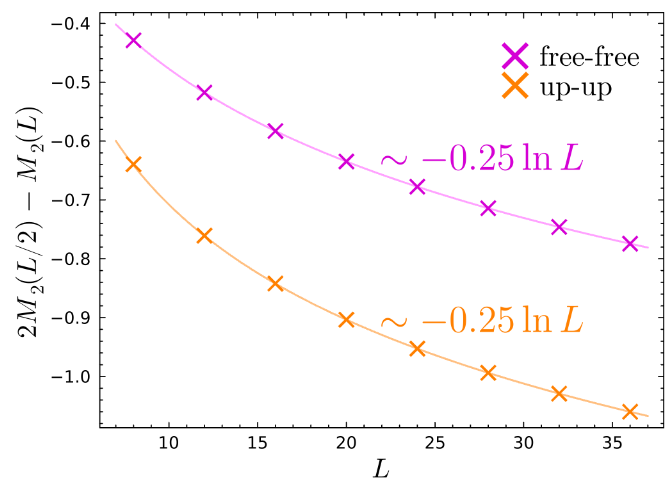
Open boundaries (totally reflective, factorising)





$$M_\alpha = m_\alpha L + \gamma_\alpha \ln L$$

logarithmic term due to the corners



Ising CFT: conformal defects move and fuse via Clifford unitaries

Cardy states $|f\rangle\,,\ \ |\uparrow\rangle\,,\ \ |\downarrow\rangle$

Verlinde lines $1,\ \eta,\ \mathcal{D}$

N. Seiberg, et al., SciPost (2024)

Fusion rules

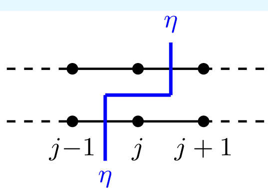
$$\eta * |f\rangle = |f\rangle\,, \qquad \eta * |\uparrow\rangle = |\downarrow\rangle\,, \qquad \eta * |\downarrow\rangle = |\uparrow\rangle\,,$$
$$\mathcal{D} * |f\rangle = |\uparrow\rangle + |\downarrow\rangle\,, \qquad \mathcal{D} * |\uparrow\rangle = |f\rangle\,, \qquad \mathcal{D} * |\downarrow\rangle = |f\rangle\,.$$

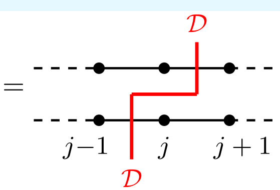
$$\eta \otimes \eta = 1, \qquad \eta \otimes \mathcal{D} = \mathcal{D} \otimes \eta = \mathcal{D},$$
$$\mathcal{D} \otimes \mathcal{D} = \mathcal{T}^- \oplus \mathcal{T}^- \eta.$$

Lattice translation defect $\mathcal{T}^-$

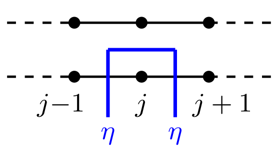
In the continuum: $\mathcal{N} \otimes \mathcal{N} = 1 \oplus \eta$

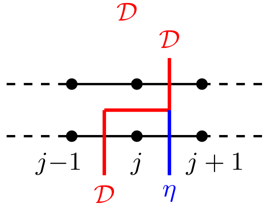
Movement operators

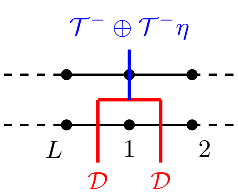
$U_\eta^j = X_j =$ 

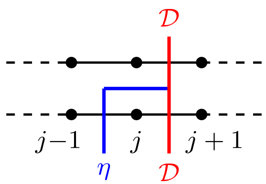
$U_{\mathcal{D}}^j = \mathsf{CZ}_{j,j+1}\mathsf{H}_j =$ 

Fusion operators

$\lambda_{\eta\otimes\eta}^j = X_j =$ 

$\lambda_{\mathcal{D}\otimes\eta}^j = U_{\mathcal{D}}^j X_j =$ 

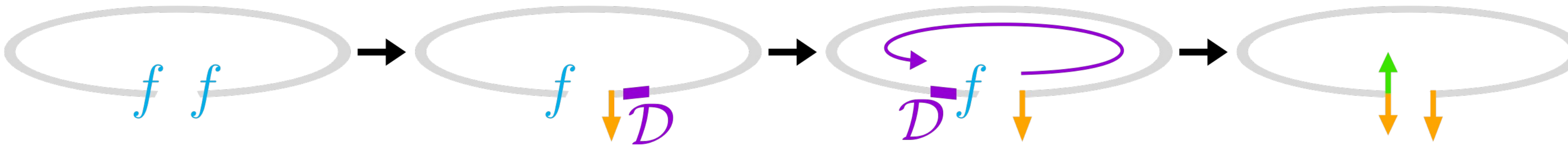
$\lambda_{\mathcal{D}\otimes\mathcal{D}}^1 = \mathsf{H}_1\mathsf{CZ}_{1,2} =$ 

$\lambda_{\eta\otimes\mathcal{D}}^j = Z_j X_{j+1} =$ 

also for fusion with boundaries

SRE encodes fusion rules

$$\mathcal{D}* |{\downarrow}\rangle = |f\rangle\,, \quad \mathcal{D}* |f\rangle = |{\uparrow}\rangle + |{\downarrow}\rangle$$



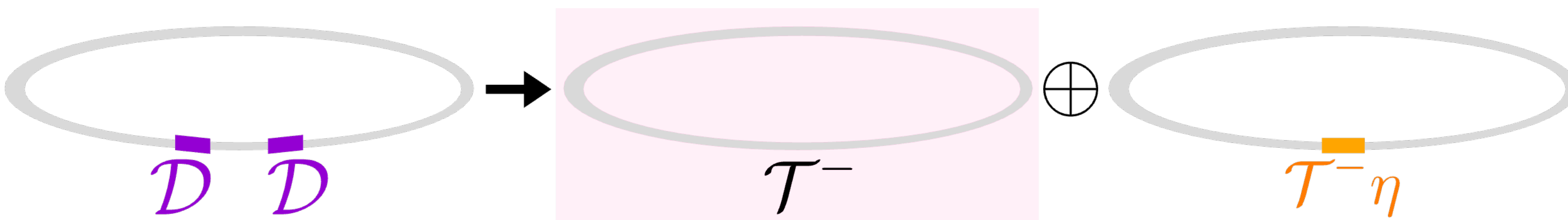
SRE with boundaries $(\,|f\rangle\,,\,|f\rangle)$ **is equal to that with** $(\,|{\uparrow}\rangle + |{\downarrow}\rangle\,,\,|{\downarrow}\rangle)$

- **reduce EE by a constant (contribution from the boundary entropy)**
- **disentangling with Clifford unitaries (CAMPS) might be useless**

also for the XXZ chain?

$$\mathcal{D} \otimes \mathcal{D} = \mathcal{T}^- \oplus \mathcal{T}^- \eta$$

ground state sector



SRE with two duality defects is equal to that with a $\mathcal{T}^-$ **defect (periodic chain with one less site)**

SRE with multiple defects is determined by the fusion rule!

Measurements as boundary perturbation

Gibbs state

$$\rho = \frac{e^{-\beta H}}{\mathrm{Tr} e^{-\beta H}}$$

expectation value

$$\langle A \rangle = \mathrm{Tr}[A\rho] = \frac{1}{Z}\int \mathcal{D}\varphi \; A[\varphi]\, e^{-\mathcal{S}[\varphi]}$$

partition function

$$Z = \int \mathcal{D}\varphi \, e^{-\mathcal{S}[\varphi]}$$

projective measurement $\{P^{\vec{m}}\}_{\vec{m}}$

post-measurement state

$$\rho^{\vec{m}} = \frac{P^{\vec{m}} \rho P^{\vec{m}}}{p_{\vec{m}}}$$

Born probability

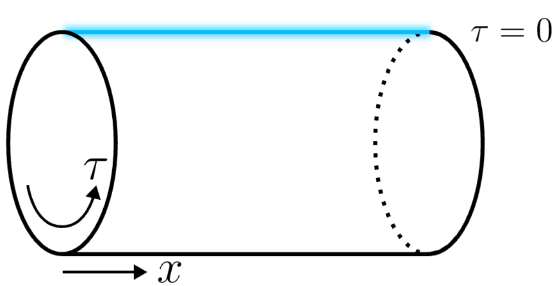
$$p_{\vec{m}} = \mathrm{Tr}[P^{\vec{m}}\rho] = \frac{1}{Z}\int \mathcal{D}\varphi \; P^{\vec{m}}[\varphi]\, e^{-\mathcal{S}[\varphi]}$$

$$= \frac{1}{Z}\int \mathcal{D}\varphi \; e^{-\mathcal{S}[\varphi]-\delta\mathcal{S}[\varphi]}$$

$$P^{\vec{m}}[\varphi] = e^{-\delta\mathcal{S}[\varphi]}$$

$$Z^{\vec{m}} = \int \mathcal{D}\varphi \, e^{-\mathcal{S}[\varphi]-\delta\mathcal{S}[\varphi]}$$

$$=$$



Boundary perturbation $\delta\mathcal{S}[\varphi]$

- **flows to conformal boundary condition in the IR limit**
- **infinitely large coupling for projective measurements**

S. J. Garratt et al., PRX (2023)

How to determine the boundary condition

participation entropy

$$\sum_{\vec{m}} p_{\vec{m}}^n = \mathrm{Tr}\left[\sum_{\vec{m}} (P^{\vec{m}})^{\otimes n} \rho^{\otimes n}\right] = \frac{1}{Z^n} \int \mathcal{D}\vec{\varphi}\, e^{-\sum_{j=1}^n \mathcal{S}[\varphi_j] - \delta\mathcal{S}_n[\vec{\varphi}]}$$

replicated theory

$$\vec{\varphi} = (\varphi_1, \varphi_2, \ldots, \varphi_n)$$

boundary perturbation

$$e^{-\delta\mathcal{S}_n[\vec{\varphi}]} = \sum_{\vec{m}} (P^{\vec{m}})^{\otimes n}$$

Example: Bell-state measurement, $n = 2$

$$\begin{aligned} P^{b_1 b_2} &= |\mathrm{Bell}^{b_1 b_2}\rangle\langle \mathrm{Bell}^{b_1 b_2}| \\ &= \frac{I + (-1)^{b_1} ZZ}{2} \frac{I + (-1)^{b_2} XX}{2} \end{aligned}$$

$$\Rightarrow \quad \sum_{b_1, b_2} (P^{b_1 b_2})^{\otimes 2} = \frac{I + ZZ^{(1)} ZZ^{(2)}}{2} \frac{I + XX^{(1)} XX^{(2)}}{2}$$

In general,

$$\begin{aligned} \sum_{b_1, b_2} (P^{b_1 b_2})^{\otimes \alpha} &= \prod_{s \in \tilde{S}^{(\alpha)}} \frac{I + s}{2} \\ &\propto \lim_{\mu \to \infty} e^{\mu \sum s} \end{aligned}$$

$$\tilde{S}^{(\alpha)} = \{\underbrace{XX^{(1)} XX^{(2)}}, \underbrace{ZZ^{(1)} ZZ^{(2)}}, \ldots, \underbrace{ZZ^{(\alpha-1)} ZZ^{(\alpha)}}\}$$

interlayer coupling

$$\Longrightarrow \quad \delta\mathcal{S}_{2\alpha} = -\mu \int d\tau\, \delta(\tau) \int dx \boxed{\sum_{s \in \tilde{S}^{(\alpha)}} s}$$

should be expressed by operators in the CFT (requires knowledge of lattice ↔ CFT correspondence)

Ising CFT: bosonization

Lattice ↔ CFT $Z \sim \sigma,\ X - \langle X \rangle \sim \varepsilon$

Bosonization

$$\sigma_1\sigma_2 = \cos\phi,$$
$$\varepsilon_1\varepsilon_2 = (\partial_\mu\phi)^2,$$
$$\varepsilon_1 + \varepsilon_2 = \cos 2\phi.$$

$$\mathcal{L} = \frac{1}{2\pi}(\partial_\mu\phi)^2, \quad \phi \sim \phi + 2\pi, \ \phi \sim -\phi$$

free-boson CFT **compactification** **$\mathbb{Z}_2$ orbifold**

$$\begin{aligned} ZZ^{(i)}ZZ^{(i+1)} &\sim \cos\phi_i \cos\phi_{i+1} \\ &= \cos(\phi_i + \phi_{i+1}) + \cos(\phi_i - \phi_{i+1}) \\ &= 2\cos(\phi_i - \phi_{i+1}), \\ XX^{(i)}XX^{(i+1)} &\sim \langle X \rangle^2[(\partial_\mu\phi_i)^2 + (\partial_\mu\phi_{i+1})^2] \end{aligned}$$

$$\delta\mathcal{S}_{2\alpha} = -\mu \int dx \sum_{i=1}^{\alpha} \cos(\phi_i - \phi_{i+1})$$

boundary perturbation of α-component CFT
(the most relevant terms)

Boundary condition (after the folding trick)

$$\Gamma_1 : \begin{cases} \sum_{i=1}^{2\alpha} \phi_i & \text{NBC} \\ \phi_1 - \phi_2 = 0 & \text{DBC} \\ \phi_2 - \phi_3 = 0 & \text{DBC} \\ \vdots & \\ \phi_{2\alpha-1} - \phi_{2\alpha} = 0 & \text{DBC} \end{cases}$$

Neumann of the "center of mass motion"
Dirichlet of the "relative motion"
+ symmetrization for the orbifold

Ising CFT: result

$$\Gamma_2 : \begin{cases} \phi_i - \phi_{i+\alpha} = 0 & \text{DBC} \\ \phi_i + \phi_{i+\alpha} & \text{NBC} \end{cases}$$

artificial boundary

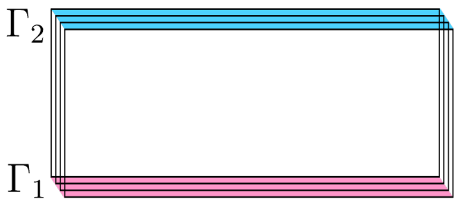
$$\Gamma_1 : \begin{cases} \sum_{i=1}^{2\alpha} \phi_i & \text{NBC} \\ \phi_1 - \phi_2 = 0 & \text{DBC} \\ \phi_2 - \phi_3 = 0 & \text{DBC} \\ \vdots & \\ \phi_{2\alpha-1} - \phi_{2\alpha} = 0 & \text{DBC} \end{cases}$$

boundary imposed by Bell-state measurements

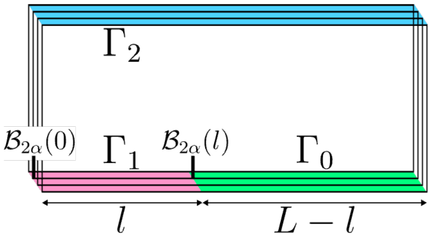
$$\Gamma_0 : \begin{cases} \phi_1 = \pi/2 & \text{DBC} \\ \phi_2 = \pi/2 & \text{DBC} \\ \vdots & \\ \phi_{2\alpha} = \pi/2 & \text{DBC} \end{cases}$$

boundary by $|\mathrm{Bell}^{00}\rangle = \frac{|00\rangle + |11\rangle}{\sqrt{2}}$

(corresponds to partial trace)



full state SRE



mutual SRE

Constant term of the full state SRE

$$g_1 = \langle \mathrm{GS}|\Gamma_1\rangle = \sqrt{\alpha} \quad \Longrightarrow \quad \boxed{c_\alpha = \frac{\ln\sqrt{\alpha}}{\alpha - 1}}$$

Coefficient of the log-scaling mutual SRE

$$\Delta_{2\alpha} = \frac{1}{16} \quad \Longrightarrow \quad \boxed{W_\alpha(l) = \frac{4\Delta_{2\alpha}}{\alpha - 1}\ln l = \frac{1}{4(\alpha-1)}\ln l}$$

$\because$ **lowest conformal weight included in the amplitude**

$$\langle \Gamma_0 | q^{L_0 + \overline{L}_0 - c/12} | \Gamma_1 \rangle = \sum_h n^h_{\Gamma_0\Gamma_1}\, \chi_h(\tilde{q}) \quad (n^h_{\Gamma_0\Gamma_1} \in \mathbb{Z}_{\geq 0})$$

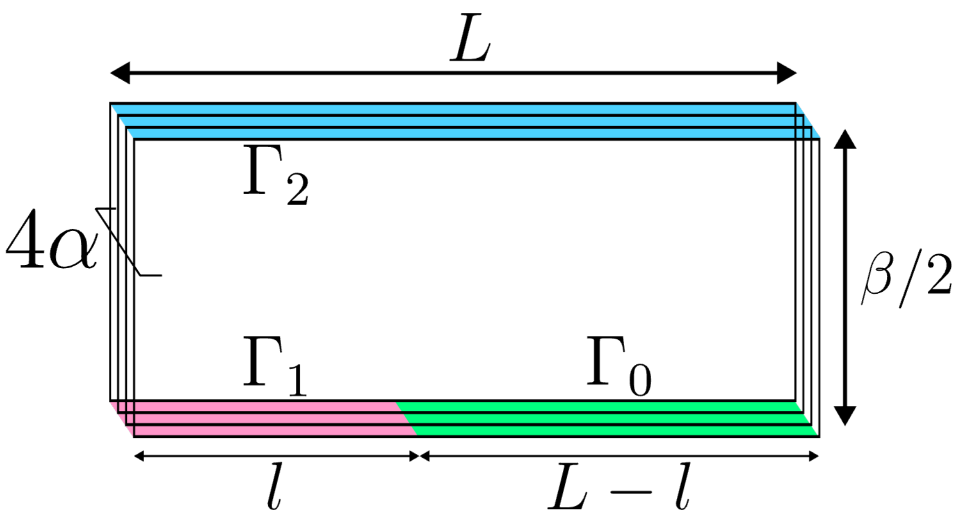
The boundary condition Γ_0

$$\mathrm{Tr}_A[\sigma^{\vec{m}_A}\rho_A] = \mathrm{Tr}\big[(\sigma^{\vec{m}_A}\otimes I_B)\psi\big] \qquad \vec{m}_A \in \{0,1\}^{2l}$$

$$\mathrm{Tr}_A^2[\sigma^{\vec{m}_A}\rho_A] = 2^L\,\mathrm{Tr}\big[(P^{\vec{m}_A}\otimes(P^{00})^{\otimes L-l})(\psi\otimes\psi^*)\big]$$

$$M_\alpha(\rho_A) = \frac{(\alpha L - l)\ln 2}{1-\alpha} + \frac{1}{1-\alpha}\ln\mathrm{Tr}\left[\sum_{\vec{m}_A}(\tilde{P}^{\vec{m}_A})^{\otimes\alpha}(\psi\otimes\psi^*)^{\otimes\alpha}\right]$$

$$\tilde{P}^{\vec{m}_A} = P^{\vec{m}_A}\otimes(P^{00})^{\otimes(L-l)}$$



Boundary condition imposed by uniform measurement of Bell00

For the Ising model, $|f\rangle\otimes|f\rangle = |D(\pi/2)\rangle_{\mathrm{orb}}$

$$\Longrightarrow \qquad |\Gamma_0\rangle = |D(\pi/2)\rangle_{\mathrm{orb}}^{\otimes 2\alpha}$$

Construction of the boundary states

Ishibashi condition

$$(L_n - \bar{L}_{-n})\,|\Gamma\rangle = 0$$

in terms of U(1) currents

$$(\vec{\alpha}^L_m - \mathcal{R}\vec{\alpha}^R_{-m})\,|\Gamma\rangle = 0$$

$\mathcal{R}$ **: N×N orthogonal matrix**

Ishibashi state (coherent state)

$$S(\mathcal{R})\,|\vec{R},\vec{K}\rangle$$

$$S(\mathcal{R}) = \exp\left[-\sum_{n=1}^{\infty}(\vec{a}^L_n)^\dagger \cdot \mathcal{R}(\vec{a}^R_n)^\dagger\right] \qquad \vec{K} + \kappa\vec{R} = \mathcal{R}(\vec{K} - \kappa\vec{R})$$

Consistent boundary state of mixed Dirichlet-Neumann boundary condition

$$|\mathcal{R}(\vec{\phi}_D,\vec{\theta}_D)\rangle := g_{\mathcal{R}} \sum_{\vec{R}\in\Lambda_{\mathcal{R}}}\sum_{\vec{K}\in\Lambda^*_{\mathcal{R}}} e^{-i\vec{R}\cdot\vec{\theta}_D - i\vec{K}\cdot\vec{\phi}_D} S(\mathcal{R})\,|\vec{R},\vec{K}\rangle$$

g-factor

Boundary state of the orbifold theory (symmetrization)

$$a = \mathrm{diag}(\pm1,\pm1,\ldots,\pm1) \in G$$

$$|G| = 2^N$$

$$G_0 = G/\{\pm I\}$$

$$|\mathcal{R}(\vec{\phi}_D,\vec{\theta}_D)\rangle_{\mathrm{orb}} = \frac{1}{\sqrt{|G|}}\sum_{a\in G} D(a)\,|\mathcal{R}(\vec{\phi}_D,\vec{\theta}_D)\rangle$$

generic

$$|\mathcal{R}(\vec{\phi}_E,\vec{\theta}_E)\rangle_{\mathrm{orb}} = \sum_{b\in G_0} D(b)\left[\frac{1}{\sqrt{|G|}}\,|\mathcal{R}(\vec{\phi}_E,\vec{\theta}_E)\rangle \pm 2^{-N/4}\,|\mathcal{R}(\vec{\phi}_E,\vec{\theta}_E)\rangle_t\right]$$

endpoint